Power Domination on Chordal Rings*

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Abstract

The power domination problem is a variant of the classical domination problem in graphs and is defined as follows. Given an undirected graph G = (V,E), the problem is to find a minimum vertex set $S_P \subseteq V$, called the power dominating set of G, such that all vertices in G are observed by the vertices of S_P. Herein, a vertex observes itself and all its neighbors, and if an observed vertex has all but one of its neighbors observed, then the remaining neighbor becomes observed as well. The minimum cardinality of a power dominating set of a graph is its power domination number. In this paper, we give upper bounds on the power domination numbers of chordal rings, and prove some exact values of power domination numbers on chordal rings .

Keywords: Power domination; Phase measurement units; Chordal Rings.

1. Introduction

Electric power companies need to monitor their system's state as defined by a set of state variables (for example, the voltage magnitude at loads and the machine phase angle at generators [19]). One method of monitoring these variables is to place phase measurement units (abbreviated PMUs) at selected locations in the system. Because of the high cost of a PMU, it is desirable to minimize their number while maintaining the ability to monitor the entire system.

The problem of locating a smallest set of PMUs to monitor the entire system is a graph theory problem closely related to the well- known vertex covering and domination problems. Let G = (V,E) be a graph representing an electric power system, where a vertex represents an electrical node (a substation bus where transmission lines, loads, and generators are connected) and an edge represents a transmission line joining two electrical nodes. A PMU measures the state variable (voltage and phase angle) for the vertex at which it is placed and its incident edges and their end vertices (these vertices and edges are said to be *observed*.) The other observation rules [11] are as follows:

1. Any vertex that is incident to an observed edge is observed.
2. Any edge joining two observed vertices is observed.
3. If a vertex is incident to a total of $k > 1$ edges and if $k - 1$ of these edges are observed, then all k of these edges are observed.

In [11], the power system monitoring problem was first studied as a variation of the well-known dominating set problem (see also [12, 13]). A set $S \subseteq V$ is a *dominating set* in G if every vertex in $V - S$ has at least one neighbor in S. The cardinality of a minimum dominating set of G is the *domination number* $\gamma(G)$. Considering the power system monitoring problem as a variation of the dominating set problem, we define a set S_p to be a *power dominating set* (abbreviated PDS) if every vertex and every edge in G is observed by S_p. The *power domination number* $\gamma_P(G)$ is the minimum cardinality of a PDS of G. A PDS of G with the minimum cardinality is called a $\gamma_p(G)$-set. Since any dominating set is a power dominating set, $1 \leq \gamma_P(G) \leq \gamma(G)$ for all graphs G. The power domination decision problem has been proven to be NP-complete [11], even when reduced to certain classes of graphs, such as bipartite graphs or chordal graphs [11], or even split graphs [16], a subclass of chordal graphs. The power domination problem is widely studied in [1-8, 10, 11, 14-17, 19-24, 26-29].

The chordal ring, CR(N, d), is a graph with vertex set $V = \{0, 1, \ldots ,N - 1\}$ and edge set $E = \{(u,v) \mid [v - u]_N = 1 \text{ or } d\}$, where $[x]_y$ denotes x modulo y. To ensure that every vertex has four adjacent vertices, we assume $d < N/2$. An example of a chordal ring with N = 13 and d = 5 is shown in Figure 1. In this paper, we design two algorithms to provide PDSs of CR(N, d) and give upper bounds of $\gamma_p(CR(N, d))$. Then we prove the exact values of $\gamma_p(CR(N, d))$, where the integers N and d satisfy some given relations.

2. Preliminaries

All graphs considered here are undirected and simple (i.e., finite, loopless, and without multiple edges). Let G = (V(G), E(G)) be a graph with vertex set V(G) and edge set E(G). The *open neighborhood* N(v) of vertex v is the set consisting of vertices

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adjacent to v, i.e., $N(v) = \{u \in V(G) \mid (u, v) \in E\}$. The *closed neighborhood* $N[v]$ of vertex v is $v \cup N(v)$.

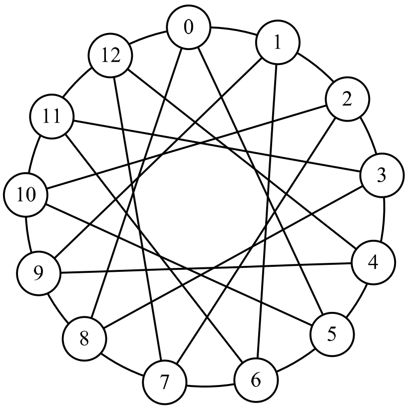


Figure 1: CR(13, 5).

In [5, 10, 15, 16], there is a way to simplify the problem description by using two rules instead of the original rules as follows:

Observation Rule 1 (OR1):

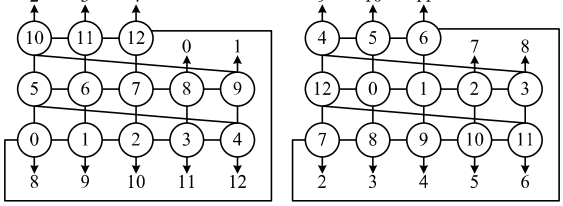
A vertex in the power domination set observes itself and all its neighbors.

Observation Rule 2 (OR2):

If an observed vertex v of degree $dg \geq 2$ is adjacent to $dg - 1$ observed vertices, then the remaining unobserved vertex becomes observed as well.

We note that OR1 is just the rule in the definition of dominating set problem. A vertex may dominate vertices at arbitrary distance when certain conditions are fulfilled by OR2 (for example, if a PMU is put in the endvertex of a path, then it can dominate all the other vertices).

Due to the regularity and the symmetry, Wong and Coppersmith [25] showed that the structure of chordal rings can be visualized in a geometrical manner such that the links (x, x + 1) and (x, x − 1) are represented by horizontal segments and the links (x, x + d) and (x, x − d) by vertical segments. We modify this representation by adding links between vertices. For example, Figure 2(a) and 2(b) show two representations of CR(13, 5).



(a) (b)

Figure 2: Two representations of CR(13, 5). The number x is pointed out by an arrow from vertex y means that there exists an edge (x, y).

3. Main Result

Lemma 1 If every vertex in consecutive 2d vertices is observed, then CR(N, d) is observed.

Proof. Chordal rings are vertex-symmetric [6]. Without loss of generality, we suppose vertex i is observed for $0 \leq i \leq 2d - 1$. Since the observed vertex d has degree 4 and its neighbors 0, d−1, d+1 are observed, by OR2, the other neighbor 2d becomes observed. Similarly, for $d+1 \leq j \leq N-d-1$, the observed vertex j has degree 4 and its neighbors j–d, j−1 and j+1 are observed, by OR2, the remaining unobserved neighbor j+d becomes observed. Consequently, all vertices in the entire CR(N, d) are observed. □

Again, since chordal rings are vertex-symmetric. Without loss of generality, we suppose first PMU is place at vertex 0, and design two algorithms to determine locations of PMUs.

Algorithm A1

Input: A CR(N, d) which $d \equiv 1$ or $d \equiv 2 \pmod 4$

Output: A power dominating set S_p of CR(N, d)

1. for $i \leftarrow 0$ to $\lceil d/4 \rceil - 1$ do
   add vertex 4i in S_p;
2. $t \leftarrow 0$;
3. If d mod 4 = 1 then $t \leftarrow 1$;
4. for $i \leftarrow \lceil d/4 \rceil$ to $\lceil d/2 \rceil - 1$ do
   add vertex 4i – t in S_p;

Algorithm A2

Input: A CR(N, d) which $d \equiv 3$ or $d \equiv 0 \pmod 4$

Output: A power dominating set S_p of CR(N, d)

for $i \leftarrow 0$ to $\lceil d/4 \rceil - 1$ do
  add vertex 4i and vertex 4i+2d – 1 in S_p;

By definition, d < N/2, so the following lemma restricts $N \geq 2d+1$.

Lemma 2 $\gamma_p(CR(N, d)) \leq \lceil d/2 \rceil$, where $d \geq 2$ and $N \geq 2d+1$.

Proof. To achieve this upper bound, we give Algorithm A1 and A2 to produce a 2PDS of CR(N, d). We now consider the following four cases in series.

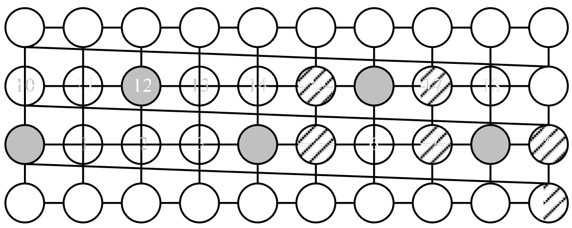
Case 1: When d mod 4 = 2 (respectively, d mod 4 = 1), by Algorithm A1, S_p = { 0, 4, 8, … , 2d–4} (respectively, S_p = {0, 4, … , d–1, d+2, d+6, … , 2d–3}). We recall that if vertex x be placed a PMU, vertices x–1, x+1, x–d and x+d are observed by OR1. It's not difficult to verify that every vertex in {N–1, 0, 1, 2, … , 2d–2} – S_p is observed by OR1. Since consecutive 2d vertices are observed, CR(N, d) is observed by Lemma 1.

Case 2: When $d \equiv 0$ or $d \equiv 3 \pmod 4$. By Algorithm A2, $S_p = \{0, 4, \ldots, 4\lceil d/4\rceil-4, 2d-1, 2d+3, \ldots, 4\lceil d/4\rceil+2d-5\}$. Vertices $x-d$, $x-1$, $x+1$ and $x+d$ are observed by OR1 where $x \in S_p$. Let $S_y = \{0, 4, \ldots, 4\lceil d/4\rceil-4\}$. For each $y \in S_y$, observed vertex $y+d-1$ has degree 4 and its neighbors $y-1$, $y+d$ and $y+2d-1$ are observed, by OR2, the other neighbor $y+d-2$ becomes observed. For each $y \in S_y$, observed vertex $y+d$ has only unobserved neighbor $y+d+1$, vertex $y+d+1$ becomes observed by OR2. Since vertex $y+d+1$ is observed, its only unobserved neighbor $y+2d+1$ becomes observed for each $y \in S_y$.

Finally, $|S_p| = \lceil d/2\rceil$ in all cases, the upper bounds on $\gamma_p(CR(N, d))$ is obtained. □

For example, Figure 3(a) shows a partial graph of CR(110, 10). Since $d \equiv 2 \pmod 4$, the PMUs are placed at vertices 0, 4, 8, 12 and 16 by Algorithm A1. Vertices 109, 1, 2, 3, 5, 6, 7, 9, 10, 11, 13, 14, 15, 17 and 18 are observed by OR1. Since every vertex in consecutive 20 vertices (from 0 to 18 and 109) are observed, then CR(110, 10) is observed by Lemma 1.

Figure 3(b) shows a partial graph of CR(72, 8). Since $d \equiv 0 \pmod 4$, the PMUs are placed at vertices 0, 4, 15 and 19 by Algorithm A2. Vertices 1, 3, 5, 7, 8, 11, 12, 14, 16, 18, 20 and 71 are observed by OR1. Since the observed vertex 7 (respectively, 8) has degree 4 and its neighbors 8, 15 and 71(respectively, 0, 7 and 16) are observed, by OR2, the other neighbor 6 (respectively, 9) becomes observed. Similarly, vertex 11 (respectively, 12) has only unobserved neighbor 10 (respectively, 13), by OR2, vertex 10 (respectively, 13) becomes observed. Finally, unobserved vertex 17 becomes observed by OR2, because it is the only unobserved neighbor of vertex 9. Since every vertex in consecutive 16 vertices (from 3 to 18) is observed, then CR(72, 8) is observed by Lemma 1.



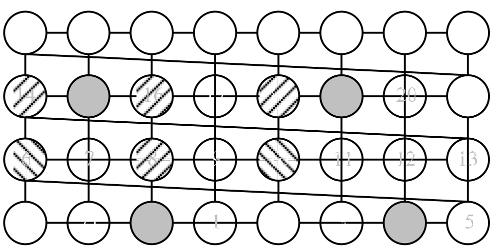


Figure 3: (a) A partial graph of CR(110, 10). (b) A partial graph of CR(72, 8). The vertex where a PMU is placed is represented by a gray vertex. The vertex is observed by OR1 (respectively, OR2) is filled with slashes (respectively, backslashes).

Lemma 3 $\gamma_p(CR(N, 2)) = 1$, where $N \geq 2d+1$.

Proof. $\gamma_p(CR(N, 2)) \geq 1$, since $\gamma_p(G) \geq 1$ for any graph G. $\gamma_p(CR(N, 2)) \leq 1$ by Lemma 2. Then $\gamma_p(CR(N, 2)) = 1$. □

Theorem 4 CR(N, d) is isomorphic to CR(N, d') if and only if dd' mod N =1. [18]

By Lemma 3 and Theorem 4, it directly derives the following corollary.

Corollary 5 $\gamma_p(CR(2d+1, d)) = 1$, where $d \geq 3$.

We note that $\gamma_p(CR(2d+1, d)) = 1$, so the following lemma restricts $N \geq 2d+2$.

Lemma 6 $\gamma_p(CR(N, d)) \geq 2$, where $d \in \{3, 4\}$ and $N \geq 2d+2$.

Proof. We will prove that CR(N, d) can not be observed by only one PMU where $d \in \{3, 4\}$ and $N \geq 2d+2$. Chordal rings are vertex-symmetric. Without loss of generality, we suppose a PMU is place at vertex 0. By OR1, vertices $N-d$, $N-1$, 1 and d are observed. Vertices 2, $d-1$ and $d+1$ are unobserved, because 2, $d-1$ and $d+1 \notin \{N-d, N-1, 1, d\}$. Vertex 1 (respectively, d) has at least two unobserved neighbors 2 and $d+1$ (respectively, $d-1$ and $d+1$). By symmetry, vertex $N-1$ (respectively, $N-d$) has 2 unobserved neighbors $N-2$ and $N-d-1$ (respectively, $N-d-1$ and $N-d+1$). Since all neighbors of vertex 0 have at least 2 unobserved neighbors, no unobserved vertex can be observed by OR2. □

By Lemmas 2 and 6, it directly derives the following corollary.

Corollary 7 $\gamma_p(CR(N, d)) = 2$, where $d \in \{3, 4\}$ and $N \geq 2d+2$.

By Corollary 7 and Theorem 4, it directly derives the following corollary.

Corollary 8 $\gamma_p(CR(N, d)) = 2$, where $d \geq 3$ and $N \in \{3d-1, 3d+1, 4d-1, 4d+1\}$.

As a concluding remark, we summarize Lemmas 2, 3 and Corollaries 5, 7, 8 as follows.

Theorem 9 The known values or upper bounds of $\gamma_p(CR(N, d))$ are shown as below.

(1) $\gamma_p(CR(N, 2)) = 1$, where $N \geq 2d+1$;
(2) $\gamma_p(CR(2d+1, d)) = 1$, where $d \geq 3$;

(3) $\gamma_p(CR(N, d)) = 2$, where $d \in \{3, 4\}$ and $N \geq 2d+2$;
(4) $\gamma_p(CR(N, d)) = 2$, where $d \geq 5$ and $N \in \{3d-1, 3d+1, 4d-1, 4d+1\}$;
(5) $\gamma_p(CR(N, d)) \leq \lceil d / 2 \rceil$, where $d \geq 5$ and $N \notin \{2d+1, 3d-1, 3d+1, 4d-1, 4d+1\}$.

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